

FOSD — Unit 6

Virtual Memory

GATE CSE PYQ source-page collection • 2017–2026

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Included source pages

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CS2019 — source page 11 — selected Q33
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Q.9 The following are some events that occur after a device controller issues an interrupt while process L is under execution.

- (P) The processor pushes the process status of L onto the control stack.
- (Q) The processor finishes the execution of the current instruction.
- (R) The processor executes the interrupt service routine.
- (S) The processor pops the process status of L from the control stack.
- (T) The processor loads the new PC value based on the interrupt.

Which one of the following is the correct order in which the events above occur?

- (A) QPTRS (B) PTRSQ (C) TRPQS (D) QTPRS

Q.10 Consider a process executing on an operating system that uses demand paging. The average time for a memory access in the system is M units if the corresponding memory page is available in memory, and D units if the memory access causes a page fault. It has been experimentally measured that the average time taken for a memory access in the process is X units.

Which one of the following is the correct expression for the page fault rate experienced by the process?

- (A) $(D - M) / (X - M)$ (B) $(X - M) / (D - M)$
(C) $(D - X) / (D - M)$ (D) $(X - M) / (D - X)$

Q.11 In an Entity-Relationship (ER) model, suppose R is a many-to-one relationship from entity set $E1$ to entity set $E2$. Assume that $E1$ and $E2$ participate totally in R and that the cardinality of $E1$ is greater than the cardinality of $E2$.

Which one of the following is true about R ?

- (A) Every entity in $E1$ is associated with exactly one entity in $E2$.
- (B) Some entity in $E1$ is associated with more than one entity in $E2$.
- (C) Every entity in $E2$ is associated with exactly one entity in $E1$.
- (D) Every entity in $E2$ is associated with at most one entity in $E1$.

- Q.32 Let the set of functional dependencies $F = \{QR \rightarrow S, R \rightarrow P, S \rightarrow Q\}$ hold on a relation schema $X = (PQRS)$. X is not in BCNF. Suppose X is decomposed into two schemas Y and Z , where $Y = (PR)$ and $Z = (QRS)$.

Consider the two statements given below.

- I. Both Y and Z are in BCNF
- II. Decomposition of X into Y and Z is dependency preserving and lossless

Which of the above statements is/are correct?

- (A) Both I and II (B) I only (C) II only (D) Neither I nor II

- Q.33 Assume that in a certain computer, the virtual addresses are 64 bits long and the physical addresses are 48 bits long. The memory is word addressable. The page size is 8 kB and the word size is 4 bytes. The Translation Look-aside Buffer (TLB) in the address translation path has 128 valid entries. At most how many distinct virtual addresses can be translated without any TLB miss?

- (A) 16×2^{10} (B) 256×2^{10} (C) 4×2^{20} (D) 8×2^{20}

- Q.34 Consider the following sets:

- S1. Set of all recursively enumerable languages over the alphabet $\{0,1\}$
- S2. Set of all syntactically valid C programs
- S3. Set of all languages over the alphabet $\{0,1\}$
- S4. Set of all non-regular languages over the alphabet $\{0,1\}$

Which of the above sets are uncountable?

- (A) S1 and S2 (B) S3 and S4 (C) S2 and S3 (D) S1 and S4

- Q.35 Consider the first order predicate formula φ :

$$\forall x [(\forall z \ z|x \Rightarrow ((z = x) \vee (z = 1))) \Rightarrow \exists w \ (w > x) \wedge (\forall z \ z|w \Rightarrow ((w = z) \vee (z = 1)))]$$

Here ' $a|b$ ' denotes that ' a divides b ', where a and b are integers. Consider the following sets:

- S1. $\{1, 2, 3, \dots, 100\}$
- S2. Set of all positive integers
- S3. Set of all integers

Which of the above sets satisfy φ ?

- (A) S1 and S2 (B) S1 and S3 (C) S2 and S3 (D) S1, S2 and S3

Consider a paging system that uses 1-level page table residing in main memory and a TLB for address translation. Each main memory access takes 100 ns and TLB lookup takes 20 ns. Each page transfer to/from the disk takes 5000 ns. Assume that the TLB hit ratio is 95%, page fault rate is 10%. Assume that for 20% of the total page faults, a dirty page has to be written back to disk before the required page is read in from disk. TLB update time is negligible. The average memory access time in ns (round off to 1 decimal places) is _____.

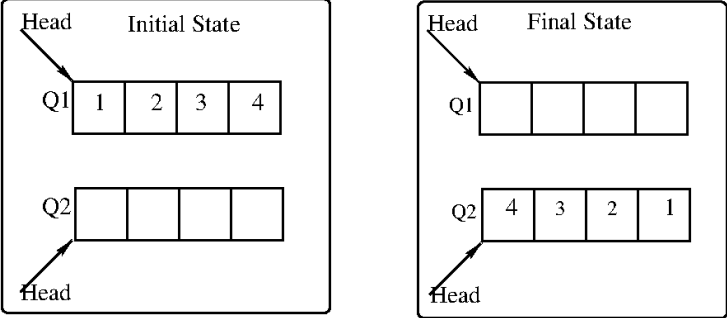
Q.No. 54 Consider a database implemented using B+ tree for file indexing and installed on a disk drive with block size of 4 KB. The size of search key is 12 bytes and the size of tree/disk pointer is 8 bytes. Assume that the database has one million records. Also assume that no node of the B+ tree and no records are present initially in main memory. Consider that each record fits into one disk block. The minimum number of disk accesses required to retrieve any record in the database is _____.

Q.No. 55 Consider a TCP connection between a client and a server with the following specifications: the round trip time is 6 ms, the size of the receiver advertised window is 50 KB, slow-start threshold at the client is 32 KB, and the maximum segment size is 2 KB. The connection is established at time $t = 0$. Assume that there are no timeouts and errors during transmission. Then the size of the congestion window (in KB) at time $t + 60$ ms after all acknowledgements are processed is _____.

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Q.38	Which one of the following statements is FALSE?
(A)	The TLB performs an associative search in parallel on all its valid entries using page number of incoming virtual address.
(B)	If the virtual address of a word given by CPU has a TLB hit, but the subsequent search for the word results in a cache miss, then the word will always be present in the main memory.
(C)	The memory access time using a given inverted page table is always same for all incoming virtual addresses.
(D)	In a system that uses hashed page tables, if two distinct virtual addresses V1 and V2 map to the same value while hashing, then the memory access time of these addresses will not be the same.
Q.39	Let $R_i(z)$ and $W_i(z)$ denote read and write operations on a data element z by a transaction T_i, respectively. Consider the schedule S with four transactions. $S: R_4(x)R_2(x)R_3(x)R_1(y)W_1(y)W_2(x)W_3(y)R_4(y)$ Which one of the following serial schedules is conflict equivalent to S?
(A)	$T_1 \rightarrow T_3 \rightarrow T_4 \rightarrow T_2$
(B)	$T_1 \rightarrow T_4 \rightarrow T_3 \rightarrow T_2$
(C)	$T_4 \rightarrow T_1 \rightarrow T_3 \rightarrow T_2$
(D)	$T_3 \rightarrow T_1 \rightarrow T_4 \rightarrow T_2$

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Q.62	Consider the queues Q_1 containing four elements and Q_2 containing none (shown as the Initial State in the figure). The only operations allowed on these two queues are Enqueue(Q, element) and Dequeue(Q). The minimum number of Enqueue operations on Q_1 required to place the elements of Q_1 in Q_2 in reverse order (shown as the Final State in the figure) without using any additional storage is _____.
	
Q.63	Consider two files systems A and B, that use contiguous allocation and linked allocation, respectively. A file of size 100 blocks is already stored in A and also in B. Now, consider inserting a new block in the middle of the file (between 50th and 51st block), whose data is already available in the memory. Assume that there are enough free blocks at the end of the file and that the file control blocks are already in memory. Let the number of disk accesses required to insert a block in the middle of the file in A and B are n_A and n_B, respectively, then the value of $n_A + n_B$ is _____.
Q.64	Consider a demand paging system with four page frames (initially empty) and LRU page replacement policy. For the following page reference string 7, 2, 7, 3, 2, 5, 3, 4, 6, 7, 7, 1, 5, 6, 1 the page fault rate, defined as the ratio of number of page faults to the number of memory accesses (rounded off to one decimal place) is _____.

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Q.57	Consider the following two-dimensional array D in the C programming language, which is stored in row-major order: <pre>int D[128][128];</pre> Demand paging is used for allocating memory and each physical page frame holds 512 elements of the array D. The Least Recently Used (LRU) page-replacement policy is used by the operating system. A total of 30 physical page frames are allocated to a process which executes the following code snippet: <pre>for (int i = 0; i < 128; i++) for (int j = 0; j < 128; j++) D[j][i] *= 10;</pre> The number of page faults generated during the execution of this code snippet is _____.

Q.60 The number of edges present in the forest generated by the DFS traversal of an undirected graph G with 100 vertices is 40. The number of connected components in G is _____

Q.61 Consider the following two regular expressions over the alphabet $\{0,1\}$:

$$r = 0^* + 1^*$$

$$s = 01^* + 10^*$$

The total number of strings of length less than or equal to 5, which are neither in r nor in s , is _____

Q.62 Consider a memory management system that uses a page size of 2 KB. Assume that both the physical and virtual addresses start from 0. Assume that the pages 0, 1, 2, and 3 are stored in the page frames 1, 3, 2, and 0, respectively. The physical address (*in decimal format*) corresponding to the virtual address 2500 (*in decimal format*) is _____

Q.63 A bag contains 10 red balls and 15 blue balls. Two balls are drawn randomly without replacement. Given that the first ball drawn is red, the probability (*rounded off to 3 decimal places*) that both balls drawn are red is _____

- Q.64 Consider a 32-bit system with 4 KB page size and page table entries of size 4 bytes each. Assume $1 \text{ KB} = 2^{10}$ bytes. The OS uses a 2-level page table for memory management, with the page table containing an outer page directory and an inner page table. The OS allocates a page for the outer page directory upon process creation. The OS uses demand paging when allocating memory for the inner page table, i.e., a page of the inner page table is allocated only if it contains at least one valid page table entry.

An active process in this system accesses 2000 unique pages during its execution, and none of the pages are swapped out to disk. After it completes the page accesses, let X denote the minimum and Y denote the maximum number of pages across the two levels of the page table of the process.

The value of $X+Y$ is _____

- Q.65 Consider the following augmented grammar, which is to be parsed with a SLR parser. The set of terminals is $\{a,b,c,d,\#, @\}$

$$\begin{aligned} S' &\rightarrow S \\ S &\rightarrow SS \mid Aa \mid bAc \mid Bc \mid bBa \\ A &\rightarrow d\# \\ B &\rightarrow @ \end{aligned}$$

Let $I_0 = \text{CLOSURE}(\{S' \rightarrow \bullet S\})$. The number of items in the set $\text{GOTO}(I_0, S)$ is _____

Q.47

Consider a demand paging system with three frames, and the following page reference string: 1 2 3 4 5 4 1 6 4 5 1 3 2. The contents of the frames are as follows initially and after each reference (from left to right):

<i>initially</i>	<i>after</i>												
-	1*	2*	3*	4*	5*	4	1	6*	4	5	1*	3*	2*
-	1	1	1	1	1	1	1	6	6	6	6	6	2
-	-	2	2	4	4	4	4	4	4	4	1	1	1
-	-	-	3	3	5	5	5	5	5	5	5	3	3

The *-marked references cause page replacements.

Which one or more of the following could be the page replacement policy/policies in use?

(A)

Least Recently Used page replacement policy

(B)

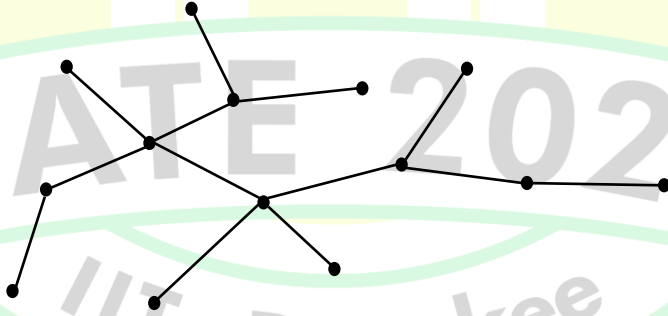
Least Frequently Used page replacement policy

(C)

Most Frequently Used page replacement policy

(D)

Optimal page replacement policy

Q.58	A computer system supports a logical address space of 2^{32} bytes. It uses two-level hierarchical paging with a page size of 4096 bytes. A logical address is divided into a b-bit index to the outer page table, an offset within the page of the inner page table, and an offset within the desired page. Each entry of the inner page table uses eight bytes. All the pages in the system have the same size. The value of b is _____. (Answer in integer)
Q.59	Consider the following algorithm <code>someAlgo</code> that takes an undirected graph G as input. <pre>someAlgo(G)</pre> 1. Let v be any vertex in G. Run BFS on G starting at v. Let u be a vertex in G at maximum distance from v as given by the BFS. 2. Run BFS on G again with u as the starting vertex. Let z be the vertex at maximum distance from u as given by the BFS. 3. Output the distance between u and z in G. The output of <code>someAlgo(T)</code> for the tree shown in the given figure is _____. (Answer in integer)
	

Q.54	In optimal page replacement algorithm, information about all future page references is available to the operating system (OS). A modification of the optimal page replacement algorithm is as follows: <i>The OS correctly predicts only up to next 4 page references (including the current page) at the time of allocating a frame to a page.</i> A process accesses the pages in the following order of page numbers: 1, 3, 2, 4, 2, 3, 1, 2, 4, 3, 1, 4. If the system has three memory frames that are initially empty, the number of page faults that will occur during execution of the process is _____. (Answer in integer)

Q.54	A system has a Translation Lookaside Buffer (TLB) that has a reach of 1 MB. TLB reach is defined as the total amount of physical memory that can be accessed through the TLB entries. The paging system uses pages of size 4 KB. The virtual address space is 64 GB and physical address space is 1 GB. If each TLB entry stores a 4-bit process id, page number, frame number, and a 2-bit control field, then the size of the TLB (in bytes) is _____. (answer in integer) <i>Note:</i> $1K=2^{10}$, $1M=2^{20}$, $1G=2^{30}$
Q.55	Consider contiguous allocation of physical memory to processes using variable partitioning scheme. Suppose there are 8 holes in the memory of sizes 20 KB, 4 KB, 25 KB, 18 KB, 7 KB, 9 KB, 15 KB, and 12 KB. Assume that no two holes are adjacent. Two processes P1 of size 16 KB and P2 of size 9 KB arrive in that order, and they are allocated memory using the best-fit technique. After allocating space to P1 and P2, the number of holes of size less than 8 KB is _____. (answer in integer) <i>Note:</i> $1K=2^{10}$
Q.56	Consider a system with 1 MB physical memory and a word length of 1 byte. The system uses a direct mapped cache, with block numbers starting from 0. The word with physical address $0xA2C28$ is mapped to the cache block number 176_{10}. The maximum possible size of the cache (in KB) for this configuration is _____. (answer in integer) <i>Note:</i> $1K=2^{10}$ and $1M=2^{20}$
Q.57	A non-pipelined instruction execution unit that operates at 1.6 GHz clock takes an average of 5 clock cycles to complete the execution of an instruction. To improve the performance, the system was pipelined with a goal of achieving an average throughput of one instruction per clock cycle. However, it could operate only at 1.2 GHz due to pipeline overheads. While executing a program in the pipelined design, 30% of instructions encountered a stall of 2 cycles due to pipeline hazards. The speed-up obtained by the pipelined design over the non-pipelined one for this program is _____. (rounded off to two decimal places) <i>Note:</i> $1G=10^9$